# Forecasting Volatility in the Chinese Stock Market: A Study Based on GARCH-Type Models and Machine Learning Algorithms

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## Abstract

This study focuses on the return series of the Shanghai Composite Index, and ChiNext Index over the past decade, using realized volatility calculated from 5-minute high-frequency data as a proxy for true volatility. To enhance prediction accuracy, the study incorporates training set residuals into machine learning algorithms such as XGBOOST, Random Forest, and GRU. Compared to the forecasting results of the GARCH model, machine learning models significantly capture volatility fluctuations during periods of high market turbulence, demonstrating notable improvements in predictive accuracy. Furthermore, by calculating utility gains, the prediction results assist investors in optimizing their investment strategies based on the mean-variance model. The findings aim to provide investors and regulatory bodies with more forward-looking and adaptive tools for anticipating market dynamics, thereby offering valuable insights for academic research in financial markets.

## CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches.

## Keywords

Machine learning, GARCH, Deep learning, Historical Moving Average

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## 1 Introduction

The studies by Bandi and Russell (2006) [1]and Hansen demonstrate that realized volatility (RV) may not serve as a consistent estimator of integrated volatility due to autocorrelation induced by market microstructure noise. Nevertheless, most literature adopts five-minute high-frequency data to mitigate the impact of market microstructure noise. The Limitations of the GARCH Model in High-Frequency Environments ,McAleer M (2008) [2] compared GARCH with the HAR-RV model, pointing out the shortcomings of GARCH in modeling long memory and asymmetry. Bandi F M (2006)[1] et al. argued that high-frequency noise distorts the estimation of volatility persistence in GARCH models. Machine Learning's Edge in Modeling Nonlinear Dependencies, Gu S (2020) [3] and Venter P J (2022) [4]. compared the performance of ML and traditional econometric methods in asset pricing, demonstrating that ML (particularly neural networks and boosted trees) can significantly improve nonlinear forecasting. Feng G (2018) [5] and Wang H (2025) [6]demonstrated that deep learning outperforms linear ARIMA and GARCH-type models in predicting asset returns. Yang Chunhui (2023) [7]similarly adopted a multi-phase forecasting approach, integrating GARCH-type models with the XGBoost algorithm, and demonstrated that this hybrid method enhanced the prediction accuracy of volatility for three stock indices. Sun Hao and Yu Bo (2020) [8] combined SVR with GARCH models for two-stage volatility forecasting, empirically confirming that this integration significantly improves volatility prediction capability. Zhu H (2023) [9] compared random forests, GARCH, and HAR models, demonstrating machine learning's superior nonlinear adaptability in high-frequency volatility forecasting. Scholars have proposed various effective volatility forecasting methods. However, due to inherent noise in financial time series and varying volatility measures' impact on prediction accuracy, using realized volatility from high-frequency data as the true volatility benchmark and applying machine learning algorithms to correct GARCH model forecast errors remains crucial for improving stock market volatility prediction.

## 2 Model Design

The specific steps are as follows: First, Calculation of Realized Volatility: Based on 5-minute high-frequency data of the Shanghai Composite Index from 2014-01-02 to 2023-12-29. Second, Estimation of the GARCH Model: Using daily closing prices of the Shanghai Composite Index from 2014-01-02 to 2021-12-31 (a total of 1,950 trading days), the daily return series is derived. A GARCH model is then fitted to this return series, and the conditional variance($\widehat{\sigma_t^2}$)is obtained using a rolling forecast method. Third, Computation of Residuals:

$$Z_{t-1} = \sigma_{t-1}^2 - \hat{\sigma}_{t-1}^2 \tag{1}$$

The squared residuals are then obtained. Then, Training the XGBOOST Model: The squared residuals ($\widehat{\varepsilon_t^2}$) are used as the feature vector.

$$Z_t = f\left(Z_{t-1}, \hat{\sigma}_{t-1}^2, \sigma_{t-1}^2\right) \tag{2}$$

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**Table 1: GARCH Model Parameter Estimation.**

| Model | Coef | Std Err | t | P > \|t\| | 95% Conf. Int. |
|---|---|---|---|---|---|
| Volatility Model | | | | | |
| omega | 0.012 | 0.007 | 1.772 | 0.076 | [-0.001, 0.025] |
| alpha [1] | 0.088 | 0.025 | 3.499 | 0.000 | [0.039, 0.138] |
| beta [1] | 0.910 | 0.023 | 40.286 | 0.000 | [0.866, 0.955] |

**Table 2: GARCH Model Parameter Estimation.**

| Model | Coef | Std Err | t | P > \|t\| | 95% Conf. Int. |
|---|---|---|---|---|---|
| Volatility Model | | | | | |
| omega | 0.035 | 0.019 | 1.810 | 0.070 | [-0.003, 0.073] |
| alpha [1] | 0.057 | 0.013 | 4.248 | 0.000 | [0.030, 0.083] |
| beta [1] | 0.933 | 0.017 | 55.859 | 0.000 | [0.901, 0.966] |

A grid search algorithm is employed to optimize the hyperparameters of the XGBOOST model, leading to the determination of the decision function $(f(\cdot))$.Last, the trained XGBOOST model is applied to predict the squared residuals for 484 trading days from 2022-01-04 to 2023-12-29. The final volatility forecasts :

$$Z_t = \hat{\sigma}_{t-1}^2 + Z_{t-1} \tag{3}$$

## 3 Volatility Clustering Test and GARCH Model Parameter Estimation

### 3.1 Shanghai Composite Index

In the table 1 above, the corresponding p-values for each parameter are all less than 0.1, which meets the requirements of the GARCH model. The model is well-fitting.

### 3.2 ChiNext Index

In the table 2 above, the estimated parameters of the variance equation, excluding the constant term, sum to a value less than 1, and the corresponding p-values for each parameter are all less than 0.1, which satisfies the requirements of the GARCH model. This indicates that the model fits well.

## 4 Forecasting Stock Market Volatility Using the GARCH-XGBoost Model

### 4.1 Shanghai Composite Index

From the Figure 1, Quantitatively, the mean squared error (MSE) of the GARCH model is 2.2885 , whereas the GARCH-XGBoost model achieves an MSE of 2.2292 , reflecting a 2.59% reduction in prediction error.

### 4.2 ChiNext Index

From the Figure 2, Compared to GARCH model's results, GARCH - XGBOOST captures volatility changes in both volatile and calm periods, with significantly improved performance. In terms of MSE, GARCH model has an MSE of 8.3117, while GARCH - XGBOOST model has an MSE of 8.977.

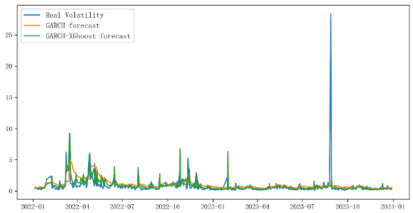



**Figure 1: SSE Index GARCH-XGBOOST Model Prediction Results.**

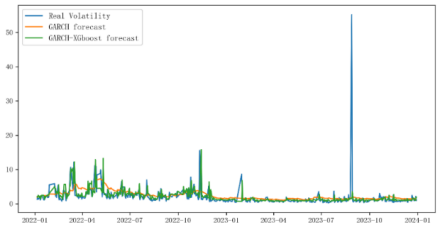



**Figure 2: ChiNext Index GARCH-XGBOOST Model Prediction Results.**

## 5 Forecasting Stock Market Volatility Using the GARCH-Random Forest Model

### 5.1 Shanghai Composite Index

From the Figure 3, compared to GARCH, GARCH – RandomForest captures volatility changes during volatile periods but overestimates it. In terms of MSE, GARCH has an MSE of 2.2885, while GARCH – XGBOOST has an MSE of 3.4182.

### 5.2 ChiNext Index

From the Figure 4, from the perspective of mean squared error (MSE), the GARCH model had an MSE of 8.3117, while the GARCH-Random-Forest model had an MSE of 9.2357.

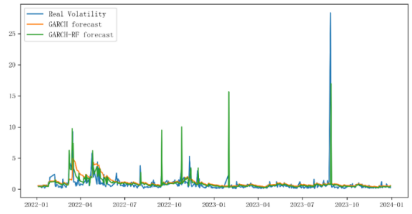



**Figure 3: Forecasting results of the Shanghai Composite Index using the GARCH-Random-Forest model.**

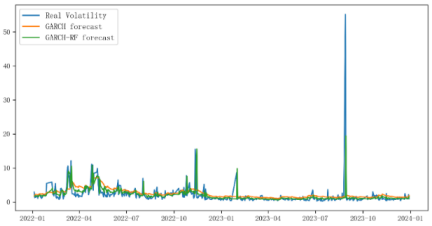



**Figure 4: Forecasting results of ChiNext Index using the GARCH-Random-Forest model.**

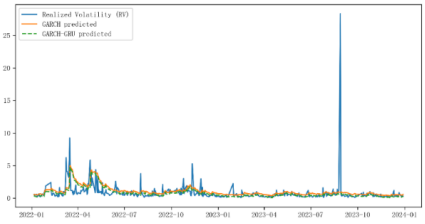



**Figure 5: Forecasting the Volatility of the SSE Index Using the GARCH-GRU Model.**

## 6 Forecasting Stock Market Volatility Using the GARCH-GRU Model

### 6.1 Shanghai Composite Index

From the Figure 5, In terms of Mean Squared Error (MSE), the GARCH model achieves an MSE of 2.2885, while the GARCH-GRU model exhibits a slightly improved MSE of 2.2517 compared to the standalone GARCH model.

### 6.2 ChiNext Index

From the Figure 6, From the perspective of Mean Squared Error (MSE), the MSE for the GARCH model is 8.3117, while the MSE for the GARCH-GRU model is 8.2010 , showing a slight reduction compared to the GARCH model.

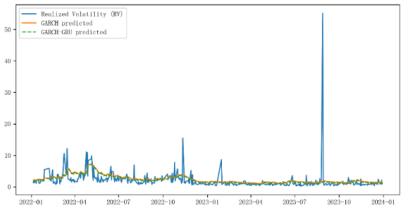



**Figure 6: Forecasting the Volatility of the ChiNext Index Using the GARCH-GRU Model.**

## 7 Model Application

The specific calculation steps are as follows:

$$w_t = \frac{1}{\gamma} \frac{\hat{y}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}} \tag{4}$$

The return prediction for time t , given information up to time t-1 , is calculated as:

$$\hat{y}_{t|t-1} = \frac{1}{win} \sum_{t-win}^{t-1} y_\tau \tag{5}$$

In the above formula, (win) represents the window size, which is set to 5. If $w_t$ is negative, it is set to 0.

$$R_{t+1} = w_t \cdot y_{t+1} + (1 - w_t) \cdot r_{f_t+1} \tag{6}$$

In this formula,$(y_{t+1})$ represents the actual index return, and$(r_{f_t+1})$represents the risk-free rate, which is approximated by the government bond yield of 1.3%. The equivalent return of the portfolio determined by the prediction model can be expressed as:

$$CER = \hat{\mu} - \frac{1}{2}\gamma\hat{\sigma}^2 \tag{7}$$

In this equation, $(\hat{\mu})$and $(\widehat{\sigma^2})$represent the simple mean and variance of the realized returns, respectively. The utility gain can be calculated as:

$$\Delta\,(\%) = 100\left(CER - CER^{(HM)}\right) \tag{8}$$

Here, $(CER^{(\mathrm{HM})})$ refers to the certainty equivalent return of the portfolio when volatility is predicted using the baseline model.

Based on the CER in the table 3, the GARCH-GRU model provides the most significant utility gain in forecasting the SSE Composite Index, suggesting that this predictive model helps investors optimize their investment strategies.

According to the the table 4, all models contribute positively to the forecasting of the ChiNext Index. indicating that this forecasting model is in a relatively optimal state for improving investors' investment strategies.

## 8 Conclusion

This paper focuses on the return series of the Shanghai Composite Index over the past decade and uses realized volatility computed from 5-minute high-frequency data as a measure of true volatility. A stage prediction method is adopted, effectively combining the GARCH model with machine learning algorithms. The high performance, flexibility, robustness, and strong generalization ability of XGBoost and GRU algorithms are fully leveraged, improving the

**Table 3: Utility Gain Results of SSE.**

| Prediction Model | $\hat{\mu}$ | $\hat{\sigma}^2$ | *CER* | $\Delta(\%)$ |
|---|---|---|---|---|
| Historical Moving Average | 1.213 | 0.024 | 1.176 | 0 |
| GARCH-XGBOOST | 1.223 | 0.016 | 1.200 | 2.4 |
| GARCH-RF | 1.234 | 0.012 | 1.218 | 4.2 |
| GARCH-GRU | 1.236 | 0.011 | 1.220 | 4.4 |

**Table 4: Utility Gain Results of ChiNext Index.**

| Prediction Model | $\hat{\mu}$ | $\hat{\sigma}^2$ | *CER* | $\Delta(\%)$ |
|---|---|---|---|---|
| Historical Moving Average | 1.244 | 0.012 | 1.226 | 0 |
| GARCH-XGBOOST | 1.250 | 0.008 | 1.238 | 1.2 |
| GARCH-RF | 1.251 | 0.007 | 1.241 | 1.5 |
| GARCH-GRU | 1.259 | 0.004 | 1.253 | 2.7 |

prediction accuracy to some extent. By calculating the utility gain, it is demonstrated that the prediction results assist investors in optimizing their investment strategies based on the mean-variance model.

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